

ASMA MUSTAFA

Data Science & Machine Learning Student | Python | ML | Deep Learning

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[GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Data Science and Machine Learning student with hands-on experience building and deploying end-to-end ML applications using Python, TensorFlow, Scikit-learn, Hugging Face Transformers, and Streamlit. Experienced in data preprocessing, feature engineering, model evaluation, deep learning, and NLP. Strong communication skills developed through experience as a Spoken English Instructor and Peer Tutor.

TECHNICAL SKILLS

- Programming: Python, OOP
- Data Science: Pandas, NumPy, Data Cleaning, EDA, Feature Engineering
- Machine Learning: Scikit-learn, Regression, Classification, Model Evaluation
- Deep Learning: TensorFlow, Keras, ANN, CNN, RNN
- NLP: Hugging Face Transformers, T5, NLP Preprocessing
- ML Deployment: Streamlit, Hugging Face Spaces
- Tools & Web: Git, GitHub, Vercel, GitHub Pages

PROJECTS

House Rent Prediction | *Python · Scikit-learn · Keras · Streamlit*

- Developed and deployed a rental-price prediction application using Linear Regression and a Keras neural network
- Improved preprocessing through target encoding for area/locality, IQR-based outlier removal, and log transformation of the target variable
- Achieved $R^2 \approx 0.80$ with the deep-learning model

Grammar & Fluency Checker (In Progress) | *Python · T5 · Hugging Face · Streamlit*

- Building a writing-assistance web app using a T5-based Hugging Face Transformers model, textstat, and difflib to assess and improve fluency
- Deployment planned on Hugging Face Spaces

Wine Quality Classification | *TensorFlow · Keras*

- Developed a neural network classifier for wine quality prediction, achieving ~87% validation accuracy

CIFAR-10 Image Classification | *TensorFlow · Hugging Face*

- Built an image classifier under a no-CNN constraint using a custom early-stopping callback
- Compared results against a Vision Transformer implementation via Hugging Face

MNIST Digit Classification | *TensorFlow · TensorFlow Datasets*

- Developed a neural-network digit classifier achieving 97–98% accuracy

Water Potability Classifier | *Scikit-learn · Streamlit*

- Built a water-potability classification model achieving ~66–68% accuracy, analyzing dataset limitations rather than overstating results

ADDITIONAL PROJECTS

- Streamlit Utility Applications: Scientific Calculator, BMI Calculator (dark-mode CSS, session history), Baby/Toddler BMI Calculator, To-Do List app with time-based overdue indicators

EXPERIENCE

Spoken English Instructor & Peer Tutor — Govt. Graduate College for Women, Lahore

- Taught spoken English and provided peer tutoring support to students
- Developed strong communication and instructional skills, now applied to explaining technical work clearly

EDUCATION

BS Computer Science — Virtual University of Pakistan

2026 – Present

Data Science and Machine Learning — NAVTTC, Lahore